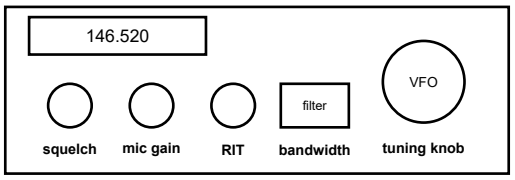
Tech Dessert #3: Radio Controls

Companion to Bite #3 -- the knobs and settings on the front of the radio, and what each one does.

Group T4B · 11 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert is the companion to Bite 3 The Bite set up the station; this sheet covers the controls you actually reach for while operating. Learn what each one does, then drill the question bank.



The controls on a typical radio's front panel. Knob positions vary by model, but every radio has these same jobs.

What Each Control Does

Control	What it does	Q
Frequency entry	The **keypad** or the **VFO knob** sets your operating frequency.	T4B02
Squelch	Mutes the receiver's hiss when no signal is present. To catch a **weak** FM signal, set the threshold so the audio is **just barely on all the time**.	T4B03
Microphone gain	Set too high on SSB, it causes **distorted** transmitted audio.	T4B01
Off-frequency FM	An FM signal received slightly off frequency sounds **distorted**.	T4B04
Scan	The scanning function **tunes through a range of frequencies**, stopping where there is activity.	T4B05
RIT (Clarifier)	Receiver Incremental Tuning. Use it when an SSB voice sounds **too high or too low** in pitch.	T4B06
Filter bandwidth	A choice of filter widths lets you **cut noise and interference** by matching the filter to the mode.	T4B08
Best SSB bandwidth	About **2400 Hz** gives the best signal-to-noise ratio for SSB voice.	T4B10
DMR code plug	**Configuration data** loaded into a DMR radio so it can reach repeaters and talkgroups.	T4B07
DMR group select	Choose a group of stations by entering the group's **identification code**.	T4B09
D-STAR setup	Your **call sign** must be programmed in before you transmit on D-STAR.	T4B11

Remember: most of these controls do one of two jobs – tuning a signal in cleanly (VFO, RIT, squelch, filter bandwidth) or sending a clean signal (microphone gain). The digital-radio settings – DMR code plugs and group codes, the D-STAR call sign – are simply data you load into the radio before you can talk.

Dessert 3 Question Bank (T4B)

T4B01. What is the effect of excessive microphone gain on SSB transmissions?
A. Frequency instability
B. Distorted transmitted audio
C. Increased SWR
D. Sideband inversion

T4B02. Which of the following can be used to enter a transceiver's operating frequency?
A. The keypad or VFO knob
B. The CTCSS or DTMF encoder
C. The Automatic Frequency Control
D. All these choices are correct

T4B03. How is squelch adjusted so that a weak FM signal can be heard?
A. Set the squelch threshold so that receiver output audio is on all the time
B. Turn up the audio level until it overcomes the squelch threshold
C. Turn on the anti-squelch function
D. Enable squelch enhancement

T4B04. What does an FM signal sound like when received slightly off frequency?
A. The audio increases in pitch
B. The audio decrease in pitch
C. There is no effect except for reduction in amplitude
D. The audio becomes distorted

T4B05. What does the scanning function of an FM transceiver do?
A. Checks incoming signal deviation
B. Prevents interference to nearby repeaters
C. Tunes through a range of frequencies to check for activity
D. Tunes through a range of frequencies to determine the antenna's resonant frequency

T4B06. Which of the following controls could be used if the voice pitch of a single-sideband signal returning to your CQ call seems too high or low?
A. The AGC or limiter
B. The bandwidth selection
C. The tone squelch
D. The RIT or Clarifier

T4B07. What is a DMR "code plug"?
A. An adapter cable used to connect a DMR radio to a computer for internet access
B. Configuration data loaded onto your radio to access repeaters and talkgroups
C. An upgrade to DMR programming software provided by the radio manufacturer to accommodate new radio models
D. A Coder-Decoder (CODEC) that converts analog voice data to DMR digital data and vice versa

T4B08. What is the advantage of having a choice of receiver filter bandwidths in a multimode transceiver?
A. Permits monitoring several modes simultaneously by selecting a separate filter for each mode
B. Permits noise or interference reduction by selecting a bandwidth matching the mode
C. Increases the number of frequencies that can be stored in memory
D. Increases the amount of offset between receive and transmit frequencies

T4B09. How is a specific group of stations selected on a DMR digital voice transceiver?
A. By retrieving the frequencies from transceiver memory
B. By enabling the group's CTCSS tone
C. By entering the group's identification code
D. By inserting a five-pin, pre-programmed code plug

T4B10. Which of the following receiver filter bandwidths provides the best signal-to-noise ratio for SSB reception?
A. 500 Hz
B. 1000 Hz
C. 2400 Hz
D. 5000 Hz

T4B11. Which of the following must be programmed into a D-STAR digital transceiver before transmitting?
A. Your call sign
B. Your output power
C. The codec type being used
D. All these choices are correct

Answer Key

T4B01: B	T4B02: A	T4B03: A	T4B04: D
T4B05: C	T4B06: D	T4B07: B	T4B08: B
T4B09: C	T4B10: C	T4B11: A	